



Bruno Ap. Silvestre Fullstack Software Engineer

CONTACT

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📄 [Currículo Lattes](#)

INFO

Age: 24 Portuguese: Proficient English: [CEFR B2](#)

SUMMARY

Fullstack Software Engineer with 4+ years of experience, nearly all of it at the same tech-education platform — which has taken me through almost every layer of it: content catalog, payment checkout, backoffice, and the infrastructure underneath. I work with Python (Django, FastAPI) on the backend, TypeScript with React, Redux, and Tailwind on the frontend, PostgreSQL and MySQL for data, and Docker to package it — and I provision what keeps it running: ECS clusters, queues, Lambdas, and CI/CD in Terraform. I own the architecture decisions on the large projects, and the areas that tolerate no silent failure — billing, credit ledgers, and authentication.

M.Sc. in Computer Science from ICMC-USP, researching adaptive assessment. The dissertation did not stop at the text: it became CAT API, an open-source system that reached production and was validated with real classrooms. That is where my interest in applied AI comes from — today I build generative AI pipelines in production, integrating LLMs and vector databases into the product.

[M.Sc. in Computer Science](#) [Interf Medal Winner](#) [SBC Marathon Finalist](#) [CEFR B2](#)

SKILLS — PROFICIENT

[Python](#) [Django](#) [FastAPI](#) [TypeScript](#) [JavaScript](#) [ReactJS](#) [Redux](#) [Vite](#) [HTML](#) [CSS](#) [Tailwind CSS](#) [styled-components](#)
[EJS](#) [SQL](#) [PostgreSQL](#) [MySQL / Aurora](#) [MariaDB](#) [Redis](#) [Celery](#) [Docker](#) [Git / GitHub](#)

CLOUD & DEVOPS

[AWS ECS](#) [AWS Lambda](#) [AWS S3](#) [CloudFront](#) [EventBridge](#) [SQS](#) [SES](#) [AWS EC2](#) [Terraform](#) [GitHub Actions](#)
[Keycloak \(SSO\)](#)

AI & DATA

[LangChain](#) [Langflow](#) [OpenAI API](#) [Qdrant](#) [Chroma](#) [Neo4j](#) [R \(mirtCAT\)](#) [Claude Code / Codex](#) [MCP](#)

SKILLS — FAMILIAR

[Java](#) [Spring](#) [Angular](#) [Android \(Java\)](#) [C](#) [C++](#) [C#](#) [Go](#) [Firebase](#)

EXPERIENCE

DIO web.dio.me Apr 2022 – Present

Mid-level Fullstack Software Engineer

Progression: Intern (Apr 2022) → Junior (Jan 2023) → Mid-level (Jul 2024)

Stack: Python (Django, FastAPI) · ReactJS + Vite + Redux + Tailwind + styled-components · PostgreSQL and MySQL (Aurora) · Docker · AWS · Terraform · Git/GitHub

Involved in every major platform project, from product to infrastructure.

KEY PROJECTS

- **DIO Play** — the platform's content catalog: courses, learning tracks, acceleration programs, live sessions, and mentorships
- **English 4 Tech** — English learning track for DIO Global users, focused on the job market
- **AI Job Hunter** — personalized AI agent that finds openings matching the candidate's résumé and tunes their profile to widen reach and improve hiring odds
- **Talent Match** — B2B platform for talent acquisition
- **Redesign of the learning components** — teaching flow and content delivery
- **Admin dashboards** — backoffice systems and internal projects

- **Automated landing pages** — a template the business team replicates for courses, learning tracks, and sales pages (Pro and Global), swapping copy and images without needing development
- **Payment checkout** — the platform's purchase flow
- Smaller projects — profile and home redesign, sales campaign configuration, among others

ENGINEERING HIGHLIGHTS

- **Technical leadership** — led architecture and structural decisions on major platform projects.
- **End-to-end backoffice** — backoffice system built from scratch: FastAPI on the backend, React + Vite + Tailwind on the frontend, including provisioning the infrastructure for both.
- **Infrastructure as code** — APIs on an ECS cluster and frontends on S3 + CloudFront, provisioned with Terraform and automated CI/CD via GitHub Actions and Docker.
- **Serverless** — Lambda deployments with Terraform and CI/CD, invoked via SQS.
- **Events and messaging** — event orchestration with EventBridge and SQS, plus asynchronous and scheduled processing with Celery Beat.
- **Caching** — cache management with Redis.
- **Payments** — checkout and integration with multiple payment providers: Malga, Pagar.me, Mercado Pago, and Barte.
- **Credit ledger** — implemented a ledger for credit control, reused across more than one feature.
- **SSO** — Keycloak deployment and configuration, integrated into both backend and frontend projects.
- **Generative AI** — pipeline with Langflow, LangChain, Qdrant, and the OpenAI API for automated content review generation.
- **Transactional email** — automated delivery via SES with templates, triggered by user events or scheduled.

Grupo MMV

Jan 2017 – Mar 2019

Hardware Technician

- **Desktop and laptop maintenance** — repairs, assembly, cleaning, formatting, and setup on Windows and Linux
- **Printers** — setup of toner and color printers
- **CCTV** — installation and configuration of cameras and DVRs
- **Network infrastructure** — rack assembly with server and switch

EDUCATION

ICMC/USP

Mar 2023 – Jul 2026

M.Sc. in Computer Science — completed

Research: AI and Education — Computerized Adaptive Testing (CAT), Item Response Theory (IRT), and Cognitive Diagnostic Models (CDM)

Dissertation: *CAT API: Exploring the Application of IRT and CDM Models in Formative Assessment Through Computerized Adaptive Testing*

■ Dissertation

📄 Completion certificate

IFSP — Araraquara Campus

Feb 2020 – Dec 2022

Systems Analysis and Development

📄 Completion certificate

PROJECTS

🔗 CAT API

2023 – 2026

Master's Research — ICMC-USP

Stack: Python + Django · adaptive service in R (mirtCAT + plumber) · custom CD-CAT · MariaDB · React + Redux + styled-components · Docker · automated CI/CD · Ubuntu web server

Learning Analytics in DIO's E-learning Platform

2021 – 2022

Undergraduate Research — IFSP

Title: *Application of Learning Analytics to Identify Aspects of Non-Completion of a Course by a Student on the Digital Innovation One Platform*

Undergraduate research applied to Digital Innovation One's (DIO) e-learning platform, a community of more than 2.4 million students. The goal was to apply Learning Analytics to understand student behavior and identify the aspects that lead to non-completion of courses and bootcamps, contributing to validating the teaching methodology and evaluating the performance of the content offered.

ACADEMIC COMPETITIONS

Maratona de Programação — SBC

2020

Finalist at the national stage

The Brazilian stage of the ICPC, organized by the Brazilian Computer Society: teams of three students solve algorithmic problems in five hours.

InterIF

2020 · 2022

3rd place in 2020 · Finalist in 2022

Programming contest held between campuses of the Federal Institutes network.

Stack: C/C++ and Python

[beecrowd profile](#)

[LeetCode profile](#)

Kangaroo Mathematics Olympiad

2016

Bronze Medal

Mathematics olympiad applied in schools across Brazil by Canguru de Matemática Brasil.

POSCOMP

2022

Score: 34

Brazil's National Exam for Admission to Graduate Programs in Computing: a nationwide assessment of Computer Science knowledge used as an admission criterion for master's and doctoral programs.